Welcome to Applied Business Statistics

Lecture 1: Introduction to Statistics

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March 3

Textbook: Keller, G. & Gaciu, N. (2020). Statistics for Management and Economics, 2nd ed., Cengage, ISBN: 9781473768260.

Before We Talk Numbers...

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Honest check-in: How do you feel right now when you hear the word "Statistics"?

Bring on the numbers! 📊

I'm cautiously optimistic. 🤔

Slightly nervous, to be honest. 😅

Is it too late to leave the room? 🏃💨

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When you have to deal with data, what is your current "weapon of choice"?

I write code (Python, R, etc.) 🐍

Excel is my absolute best friend 📗

Basic spreadsheets, if I really have to 🤷‍♀️

A calculator and a piece of paper 📝

I actively run away from data! 🙈

Course Overview: From Data to Inference

Part 1: Descriptive Statistics (Lectures 1–4)

Focus: Mastering the Python "Cookbook" and data visualization.

- **L1-L2:** Introduction to Python, Pandas, and Data Types.
- **L3-L4:** Numerical measures (Mean, SD) and Data Quality/Sampling.
- **Milestone:** By L4, you will be able to work **independently in Python** to describe any dataset.

Part 2: Probability Theory (Lectures 5–8)

Focus: Understanding the "Math" of uncertainty.

- **L5-L6:** Probability rules and Discrete Random Variables.
- **L7-L8:** Continuous Distributions (The Normal Distribution) and Sampling Distributions.
- **Insight:** Learning how samples behave compared to the population.

Part 3: Statistical Inference (Lectures 9–10)

Focus: Making business decisions based on data.

- **L9:** Estimation and Confidence Intervals.
- **L10:** Review and Final Exam Preparation.
- **Outcome:** Drawing conclusions about a population based on a sample.

Exam: What You Need to Know

Course: Applied Statistics (Spring 2026)

- **Date & Time:** Wednesday, 10 June 2026 | 09:00 – 13:00 (4 Hours)
- **Location:** ....
- **Format:** Written individual exam
- **Assessment:** Danish 7-point grading scale

Permitted Aids (The "Open Book" Policy)

- **Resources:** Textbook (Keller & Gaciu), lecture slides, personal notes, and Python notebooks.
- **Software:** Google Colab (must be used in **offline mode**; no external browsing).
- **Strictly Prohibited:** Active internet connection, communication with others, and **GenAI tools** (ChatGPT, Claude, etc.).

Warning: You must be able to write and execute Python code independently. Use GenAI during the semester to learn the logic, not to bypass the struggle of coding.

Exam Content & Competencies - What will you be tested on?

Structure of the Exam

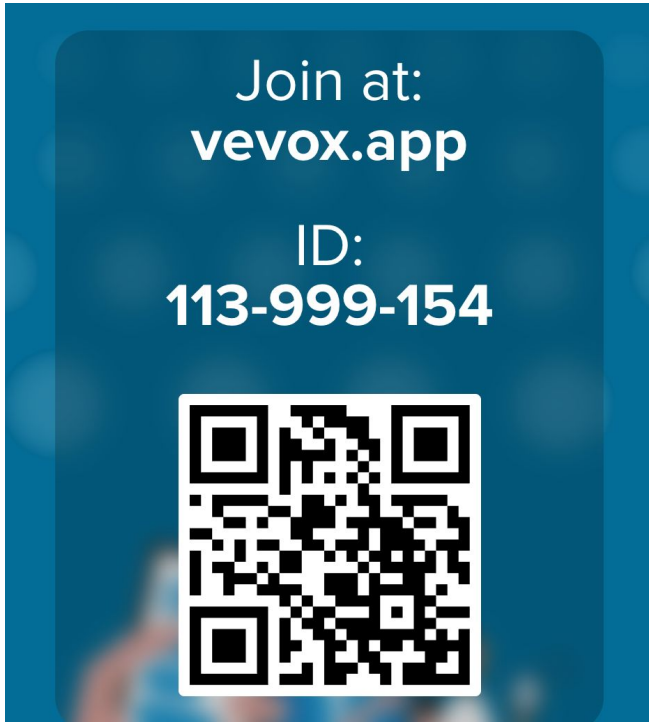
- **Conceptual (Multiple Choice):** Testing your "Statistical Imagination" and understanding of core definitions (e.g., Population vs. Sample).
- **Applied (Problem Solving):** Using Python to load datasets, calculate numerical descriptives, and generate visualizations.
- **Interpretive (Discussion):** Explaining *why* a specific method was chosen and what the results mean in a business economics context.

Key Skills Required for Success

1. **Method Selection:** Identifying the right tool for the data type (L2–L4).
2. **Inference:** Drawing valid conclusions about a population from sample statistics (L9–L10).
3. **Communication:** Presenting findings clearly and honestly, avoiding common pitfalls like "Correlation vs. Causality."
4. **Python Proficiency:** Navigating your "Python Cookbook" to solve problems without external AI assistance.

Why should I learn Statistics? And how is this course actually important for me?

What do you think is the most valuable asset in the world today?



The Economist
MAY 6TH–12TH 2017
Crunch time in France
Ten years on: banking after the crisis
South Korea's unfinished revolution
Biology, but without the cells
The world's most valuable resource
Data and the new rules of competition
Google
amazon
UBER
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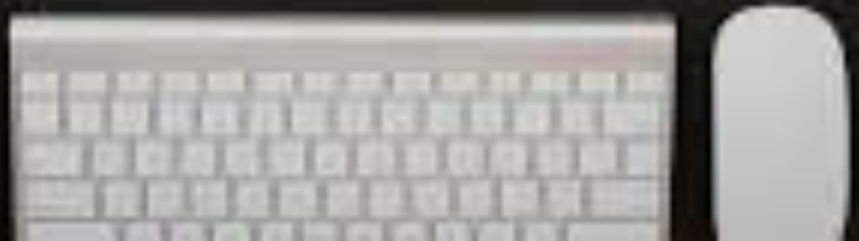
6.3 MILLION SEARCHES PER MINUTE



The Mystery Numbers Game

7, 4, 02, 10, 12, 7, -3, 4, 7, 00, 10, 7, 02, 4, 12, 7, 4, 00,

7, 10, 02, 4, 7, 12

What do these numbers mean to you right now?

The Mystery Numbers Game

7, 4, 02, 10, 12, 7, -3, 4, 7, 00, 10, 7, 02, 4, 12, 7, 4, 00,

7, 10, 02, 4, 7, 12

Final Exam Grades - Last Year -------> Data

Is this exam hard to pass?

Let's look at the data. You have 5 minutes...

Did last year's class do well?

7, 4, 02, 10, 12, 7, -3, 4, 7, 00, 10, 7, 02, 4, 12, 7, 4, 00,

7, 10, 02, 4, 7, 12

Mine, baby, mine!

Let us mine data and extract information from it!

The Question We Asked	The Information We Gained	The Statistical Tool
"How did the class perform overall?"	**Result: 5.75** The overall performance sits between a 4 (fair) and a 7 (good). The exam was balanced—not too easy, not too brutal.	**The Mean** (Average)
"What was the most typical result?"	**Result: 7** The "typical" student walked away with a good, solid understanding of the material.	**The Mode** (Most Frequent)
"What was the worst-case scenario?"	**Result: -3** The harsh reality: at least one student completely failed the exam and struggled significantly.	**The Minimum** (Lowest Score)
"Was the exam impossible to ace?"	**Result: 12** No, the highest standard of outstanding performance was absolutely achievable.	**The Maximum** (Highest Score)

Goals Scored Over the Last 7 Games

1 3 4 6 6 7 8

mean 5
average

mode 6
most common

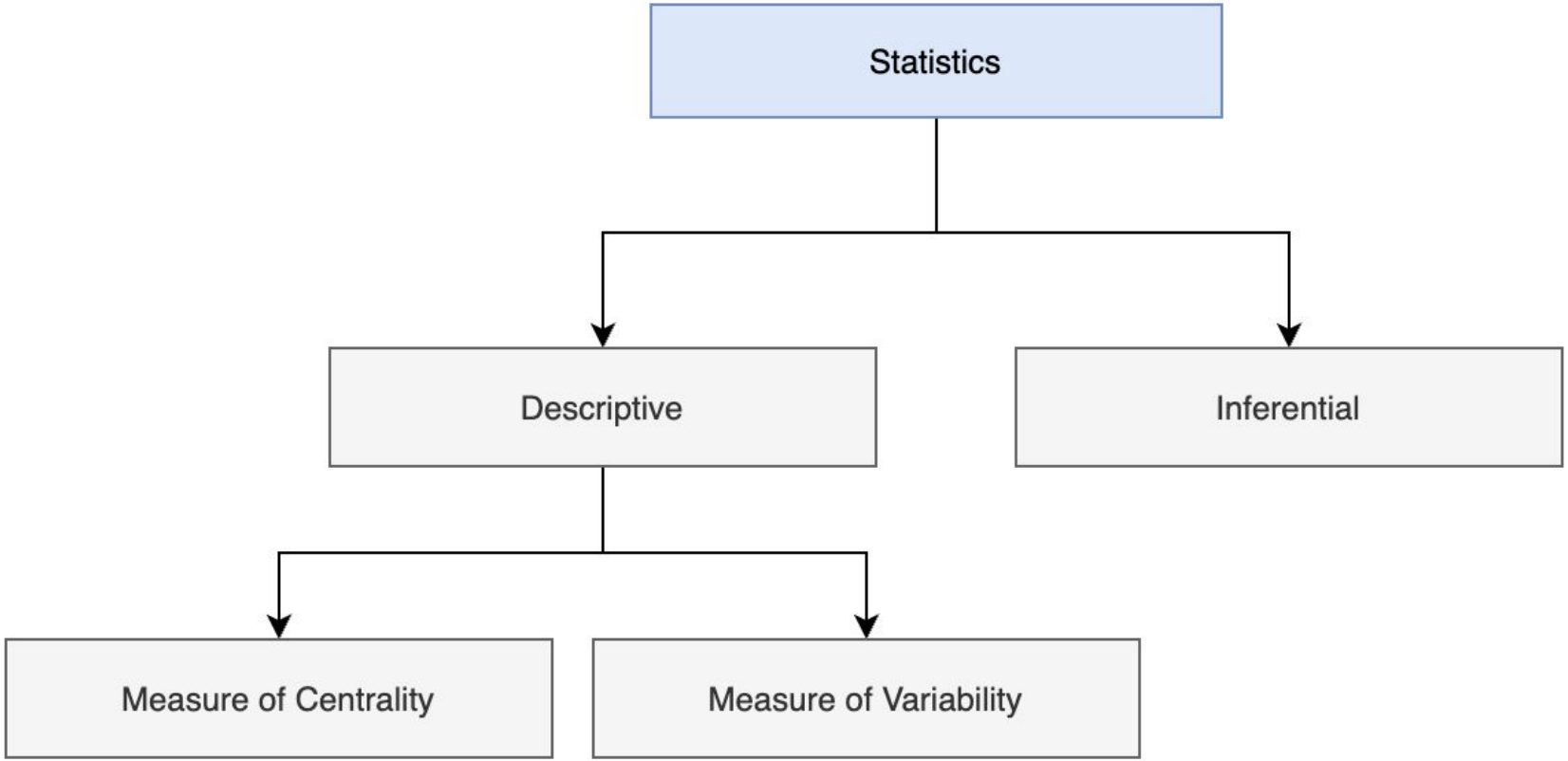
median 6
middle

range 7
largest - smallest

What is Statistics?

Data: Facts, especially numerical facts, collected together for reference or information.
Information: Knowledge communicated concerning some particular fact.

Statistics is a **tool** for creating **new understanding** from a set of numbers.


Statistics
Descriptive
Inferential
Measure of Centrality
Measure of Variability

Danish PM calls an early election seeking 'Greenland bounce'

Mette Frederiksen hopes to profit from her stand against Donald Trump's attempt to claim the Arctic territory

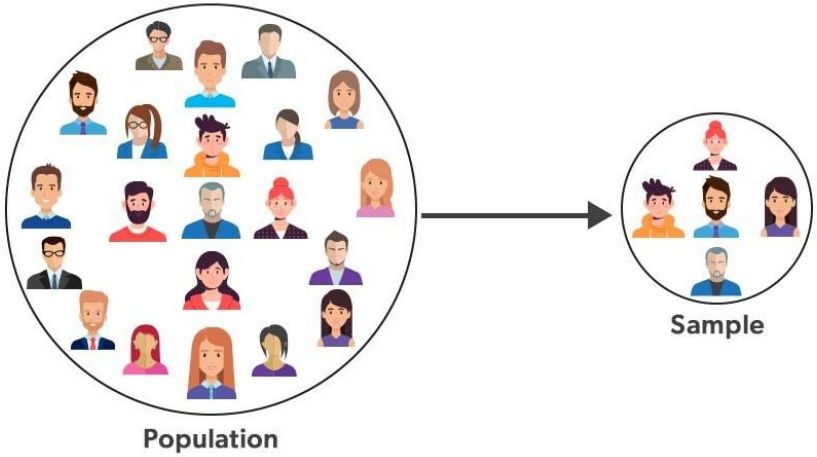
Imagine you want to predict the outcome of the next Danish election.

Calling all 4 million voters would be one approach, but this is impractical and infeasible

The Power of Sampling

Population: The entire group you want to conclude something about (e.g., all 4 million Danish voters).

Sample: The specific group of individuals that you will actually collect data from.



The Goal (Inference): We use the data from the **Sample** to make an educated guess about the **Population**. This is the essence of "Inferential Statistics" mentioned in your second slide.

Sample

A smaller group selected from the population

Eg: surveying 500 students to understand the opinions of 50,000

POPULATION

POPULATION

EXAMPLE 1: A BIASED SAMPLE (NOT REPRESENTATIVE)

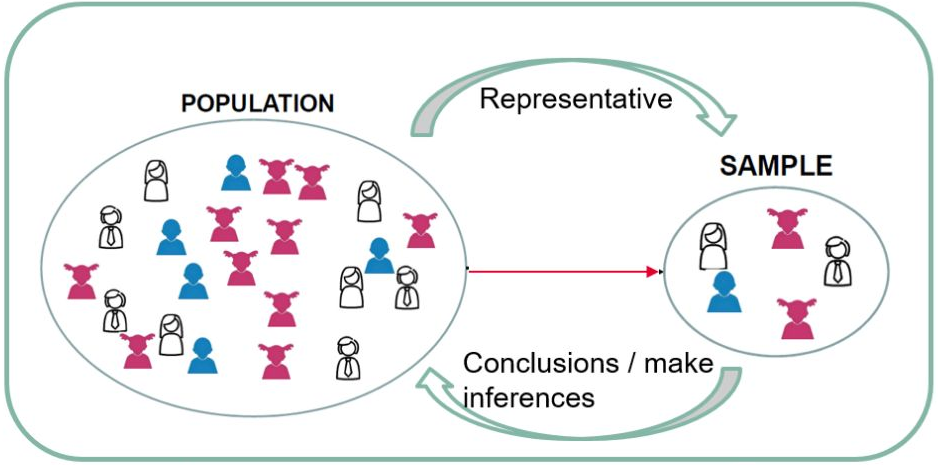
Only males were selected.
This sample does not mirror the diversity of the entire population.

EXAMPLE 3: AN IDEAL SAMPLE (RANDOM & REPRESENTATIVE)

Every member of the population had an equal chance of being selected, ensuring true representativeness.

TEACHING TIP: A good sample is a manageable group that accurately reflects the characteristics of the entire population.

Types of Statistics



Inferential Statistic

Data gathered from a sample and the statistics generated to reach conclusions about the population

Generalization, Prediction

Descriptive Statistics

Data gathered about a group (sample or population) to reach conclusions about the same group. Ex. Mean, Median …..

summarization of data

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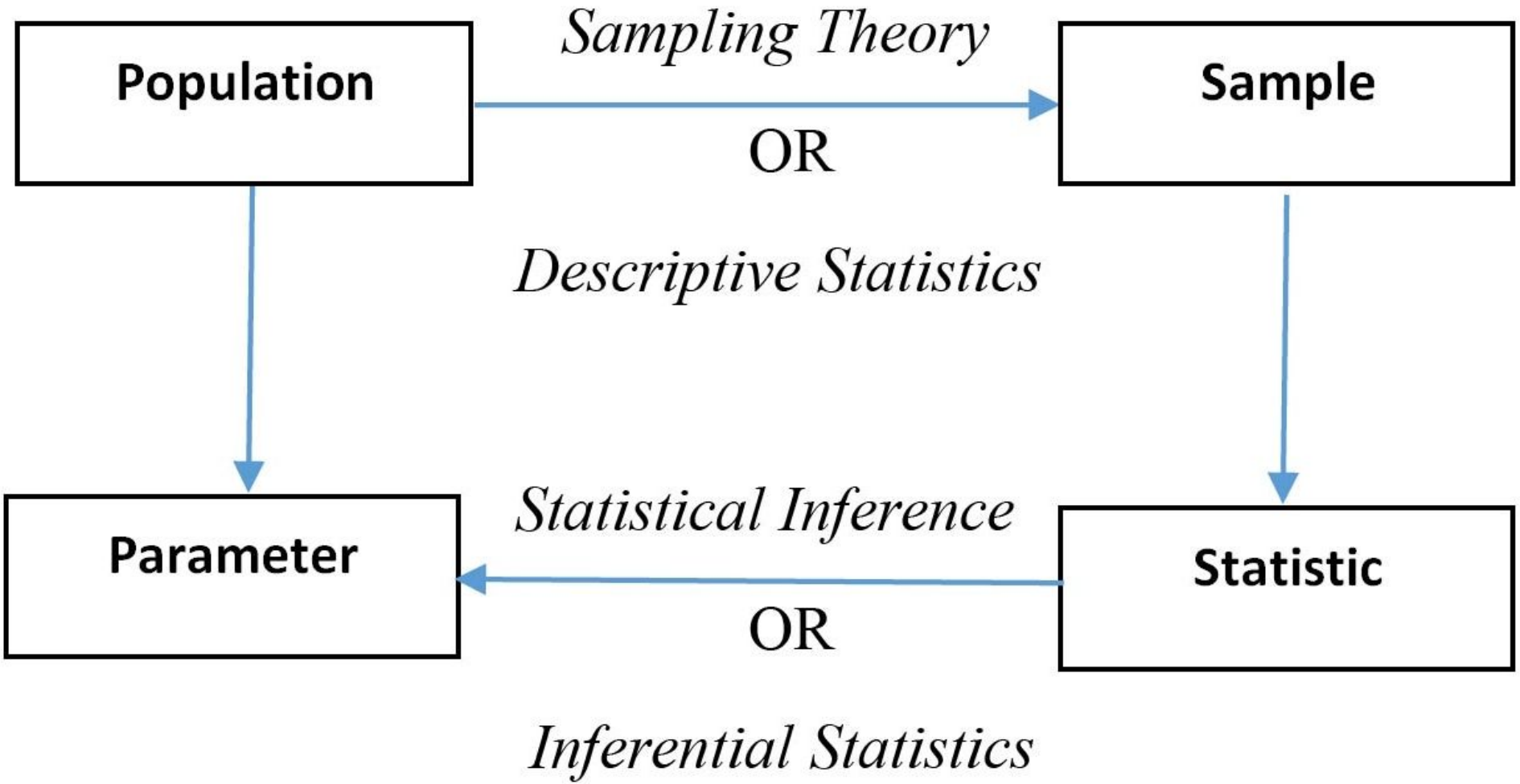
Population
Sampling Theory
Sample
OR
Descriptive Statistics
Parameter
Statistical Inference
Statistic
OR
Inferential Statistics

Characteristic	Population Parameter	Sample Statistic
Mean	μ	$\bar{X}$
Median	η	$\tilde{X}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	π (or P)	p (or $\hat{p}$)
Correlation	ρ	r
Regression Coeff.	β	b

Element	Definition	Real-World Role
Population	The "Big Picture"	What we want to understand.
Sample	The "Snapshot"	What we actually have data for.
Parameter	The "Hidden Truth"	The value we are trying to guess.
Statistic	The "Calculated Fact"	What Python tells us after running code.

The "Hidden" Wealth of Gen Z and Y

The Scenario: A bank wants to know the average savings account balance of all 18–25-year-olds in Denmark to design a new investment app.

Population (N): All ~550,000 residents in Denmark aged 18–25.

Sample (n): 1,200 students randomly surveyed across AAU, KU, and Aarhus University.

Parameter (μ): The actual average balance of all 550,000 people (a number the bank doesn't truly know).

Statistic ($\bar{x}$): The average balance of the 1,200 surveyed students (e.g., 14,500 DKK).

The Twist: If the sample only includes university students, the Statistic might be higher than the Parameter because it misses young professionals or those unemployed.

Case 2: The Mercedes EV Winter Range Test

The Story: Mercedes-Benz Denmark wants to advertise the real-world winter battery range of their new electric EQE model. There are 15,000 EQEs on the road across the country. To get their numbers, they pull telemetry data from a sample of 500 EQEs owned by a corporate fleet that operates exclusively within central Copenhagen.

Question 1: What is the Parameter (μ) we are trying to estimate?

Question 2: If the Sample Mean ($\bar{x}$) is 450 km per charge, is this a "Fair" statistic?

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Question 1: What is the Parameter (μ) we are trying to estimate?

Answer: The true average winter battery range for *all* 15,000 Mercedes EQEs driving in all conditions across Denmark.

Question 2: If the Sample Mean ($\bar{x}$) is 450 km per charge, is this a "Fair" statistic?

Answer: No. This is a classic example of Sampling Bias (specifically, selection bias based on geography and usage).